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CardioPredictor: A Real-time Non-Invasive and Cuffless Cardiovascular Disease Prediction System using 10-Fold Cross Validation on The Framingham Dataset

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Abstract & Introduction

CardioPredictor is an innovative, non-invasive, and cuffless health monitoring device to measure inflammatory biomarkers like glucose, cholesterol, and blood pressure using low-cost sensors. Data is displayed on an Organic Light Emitting Diode (OLED) screen and sent to a ThingSpeak public server within 30 seconds. A web-based predictive model, harnessing the predictive power of Gradient Boosting method, predicts the 10-year probability of developing cardiovascular disease (CVD) based on the Framingham dataset[1]. With 10-fold cross-validation, the model obtained 89% accuracy. This solution allows for fast and easy access to health insights that feeds into a sustainable healthcare system.

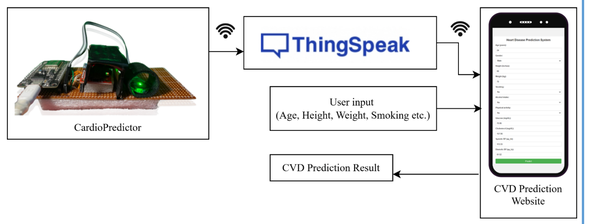


Fig. 1. System block diagram of the CardioPredictor system

Objectives

1. To develop a low-cost, non-invasive device to measure glucose, cholesterol, and blood pressure with real-time data display and cloud monitoring [2][3].
2. To provide 10-year CVD risk prediction using a web-based Gradient Boosting model with 89% accuracy[4].

Circuit Diagram

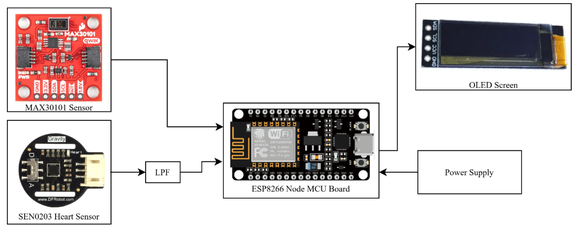


Fig 2. Connection diagram of the CardioPredictor system

Methodology

Table 1. Required Data for CVD Prediction System

SL No.	Data	Specification
01	Glucose Level	mg/dl
02	Cholesterol Level	mg/dl
03	Systolic Blood Pressure (SBP)	mmHg
04	Diastolic Blood Pressure (DBP)	mmHg
05	Heart Rate	BPM
06	Age	days
07	Height	cm
08	Weight	Kg
09	Gender	0 = Male; 1 = Female
10	Smoking habit	0 = No; 1= Yes
11	Alcohol intake	0 = No; 1= Yes
12	Physical activity	0 = Inactive; 1= Active

Glucose (mg/dL) = -0.0011*X* + 194.87 .....(*i*)
Cholesterol (mg/dL) = -0.0033*X* + 531.49.....(*ii*)
here, X is the average value of IR

SBP (mmHg) = 184.3 - 1.329 x HR(bpm) +0.00848 x Td ....(*iii*)
DBP (mmHg = 55.96 - 0.02912 x HR(bpm) +0.02302 x Td...(*iv*)
here, Td = HR(ms) – Time Delay; Time Delay = HR(bpm)+ PTT

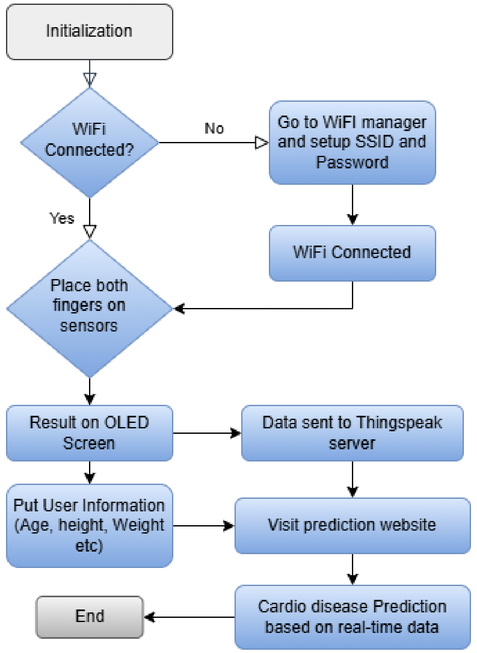


Fig 3. Total workflow of the system

Result



Fig 4. Data on OLED Screen

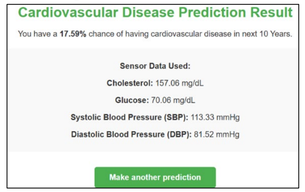


Fig 6: CVD Prediction result on webpage

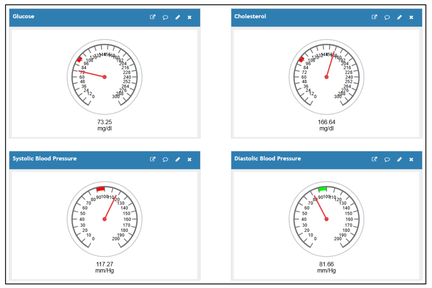


Fig 5. Data sent on Thingspeak Server

Table 2. SBP and DBP Measurement Comparison (CardioPredictor Vs Digital ARM Blood Pressure Monitoring Device)

Statistic	SBP – Cardio-Predictor (mmHg)	SBP – Digital (mmHg)	DBP – Cardio-Predictor (mmHg)	DBP – Digital (mmHg)
Mean	117.29	125.35	79.69	85.85
Standard Deviation	11.09	16.67	8.04	8.33
Min Value	96.48	101.45	64.65	68.70
Max Value	142.23	151.30	99.71	105.18

Model Accuracy

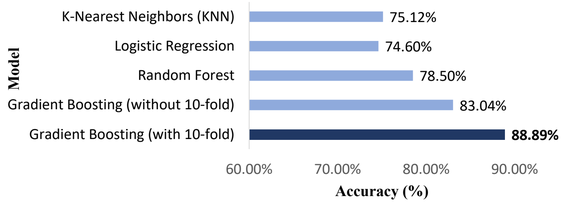


Fig 7. Model Performance comparison with different ML Models

Limitations & Future Work

1. The non-medically approved sensors have inherent noise and are affected by individual differences in finger skin depth and bone density[2].
2. The machine learning model may show demographic bias as it is trained on datasets from a Western population[1].
3. Future work includes moving to PCB design, developing a wearable device, expanding data collection from the Bangladeshi population, and conducting clinical trials for validation.

Conclusion

This work highlights the integration of non-invasive sensor technology and machine learning to provide rapid health assessments. Despite certain limitations, the system offers a practical approach for monitoring key health metrics and predicting CVD risk, making it a valuable tool for early health intervention.

References

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